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Designing Artificial Intelligence Products

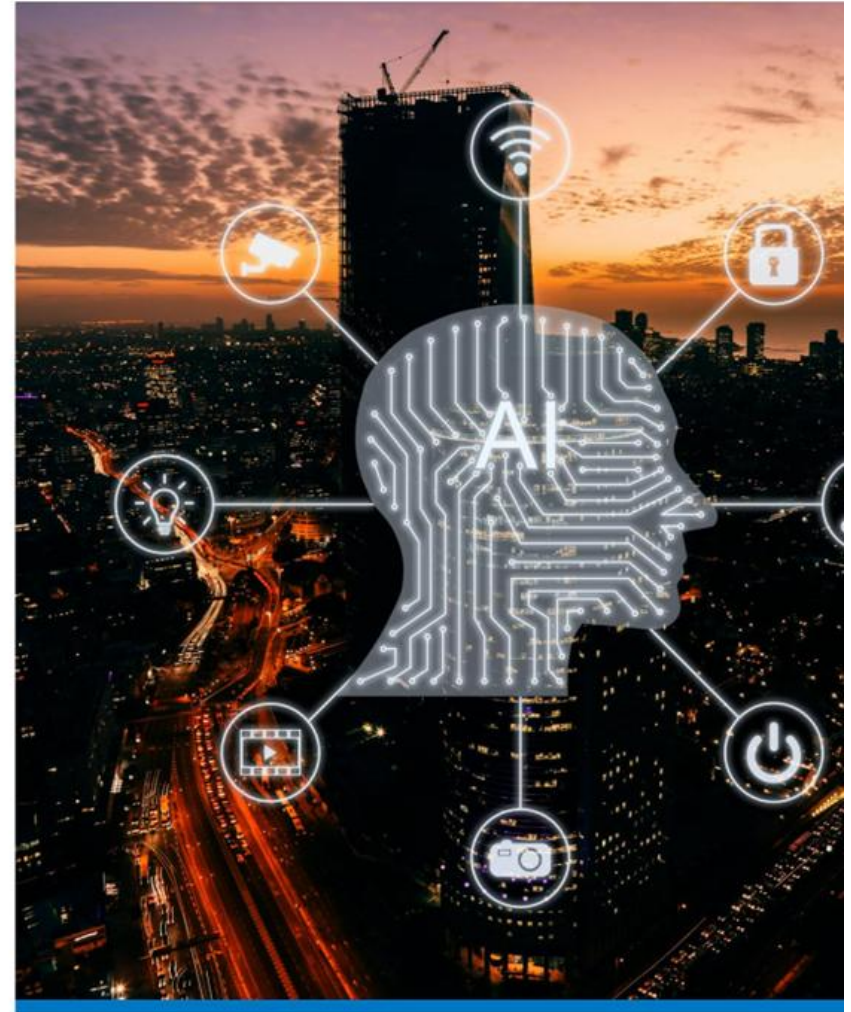
Module 1: Introduction to the Artificial
Intelligence Design Process

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Learning Outcomes

- Identify the stages of the AI design process model
- Evaluate natural language processing (NLP) applications for optimal results
- Analyze the data strategy that can yield value and sustainable advantage for an AI product or process
- Evaluate the cost metrics and contingency planning of an AI software development plan
- Describe how AI cancers affect AI applications
- Describe an AI approach to address a business problem



Artificial Intelligence (AI)

- Branch of computer science focused on replicating human-like intelligence.
- Involves cognitive functions: understanding language, learning, reasoning, problem-solving.
- Integrates fields like machine learning, neuroscience, and data science.

Machine Learning (ML)

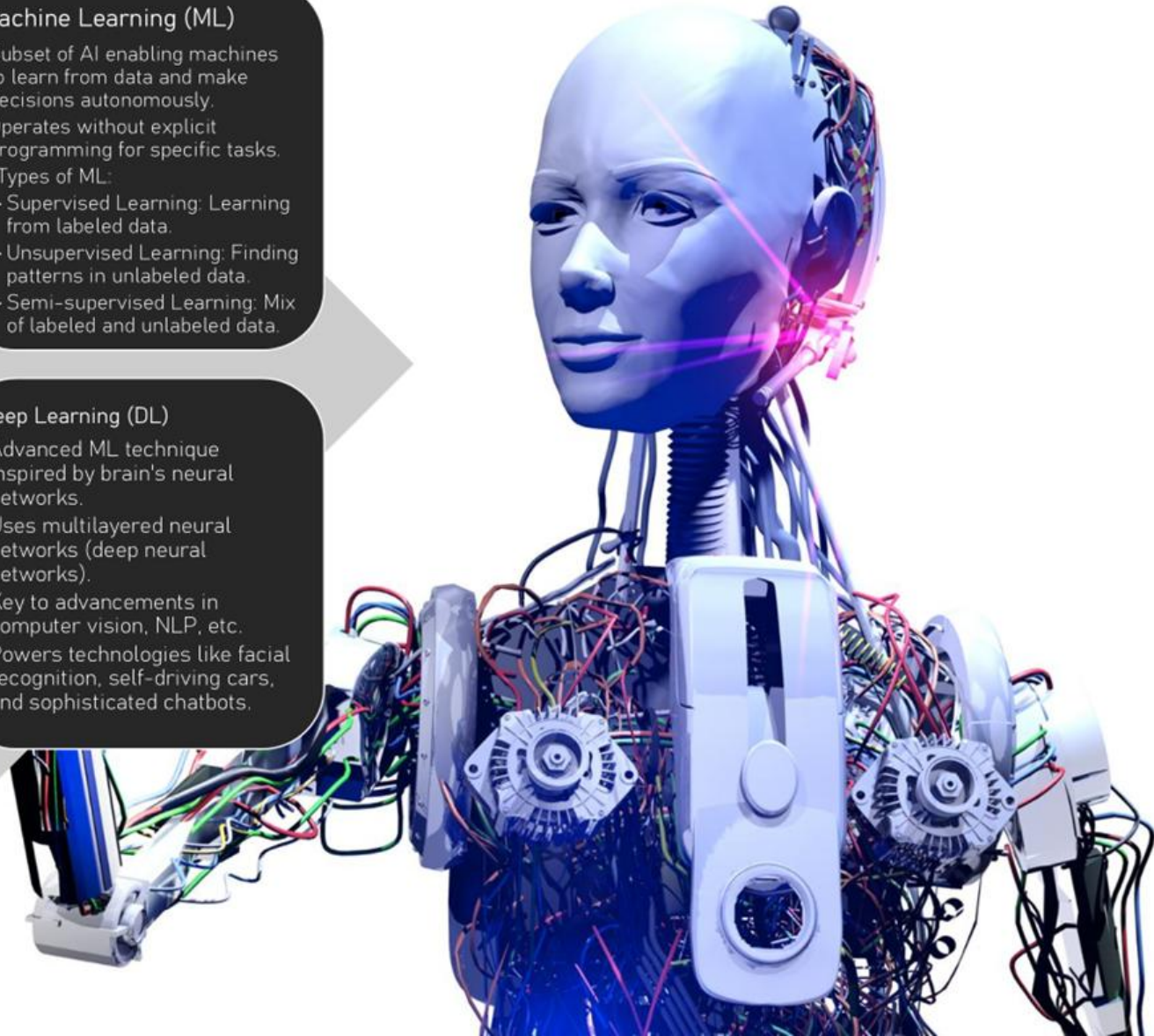
- Subset of AI enabling machines to learn from data and make decisions autonomously.
- Operates without explicit programming for specific tasks.
- Types of ML:
 - Supervised Learning: Learning from labeled data.
 - Unsupervised Learning: Finding patterns in unlabeled data.
 - Semi-supervised Learning: Mix of labeled and unlabeled data.

Natural Language Processing (NLP)

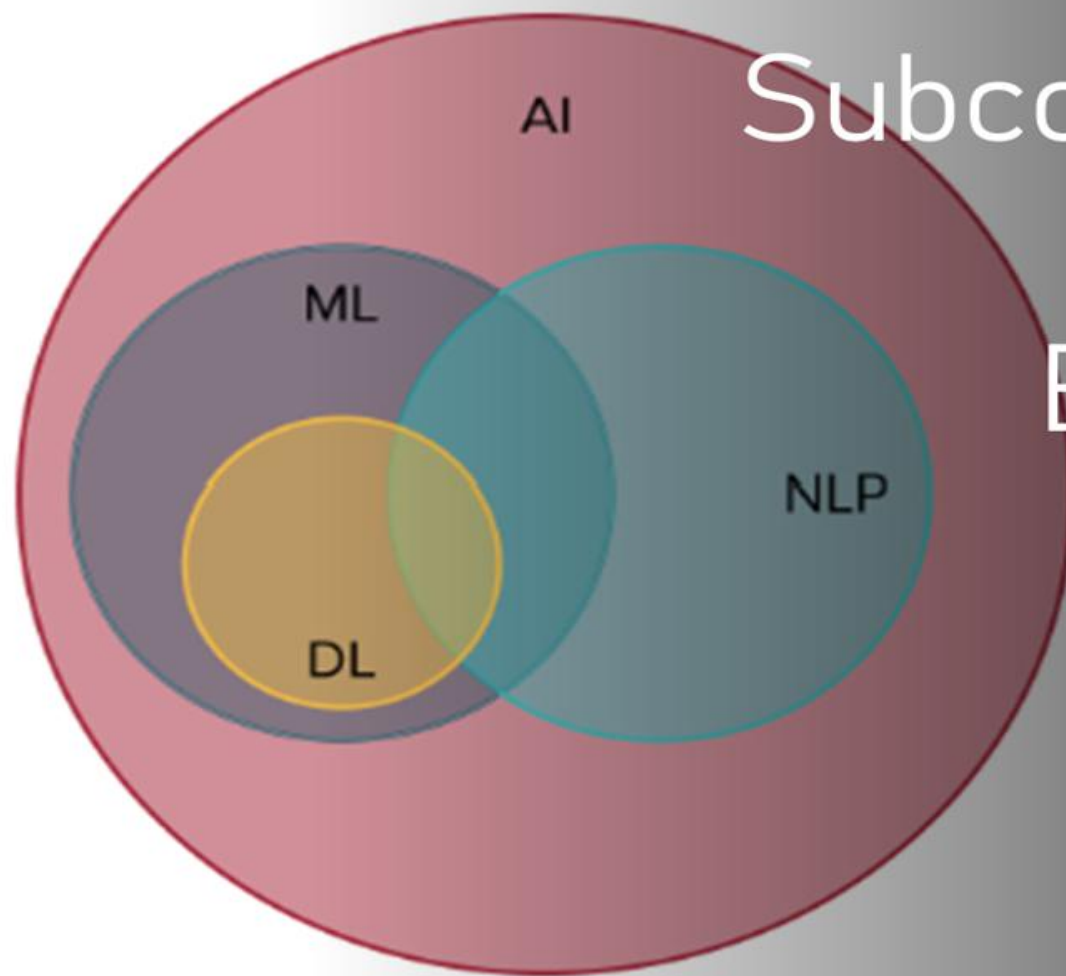
- AI field focused on interaction between computers and human language.
- Used in applications like translation, chatbots, and email filtering.
- Analyzes and derives meaning from textual data.

Deep Learning (DL)

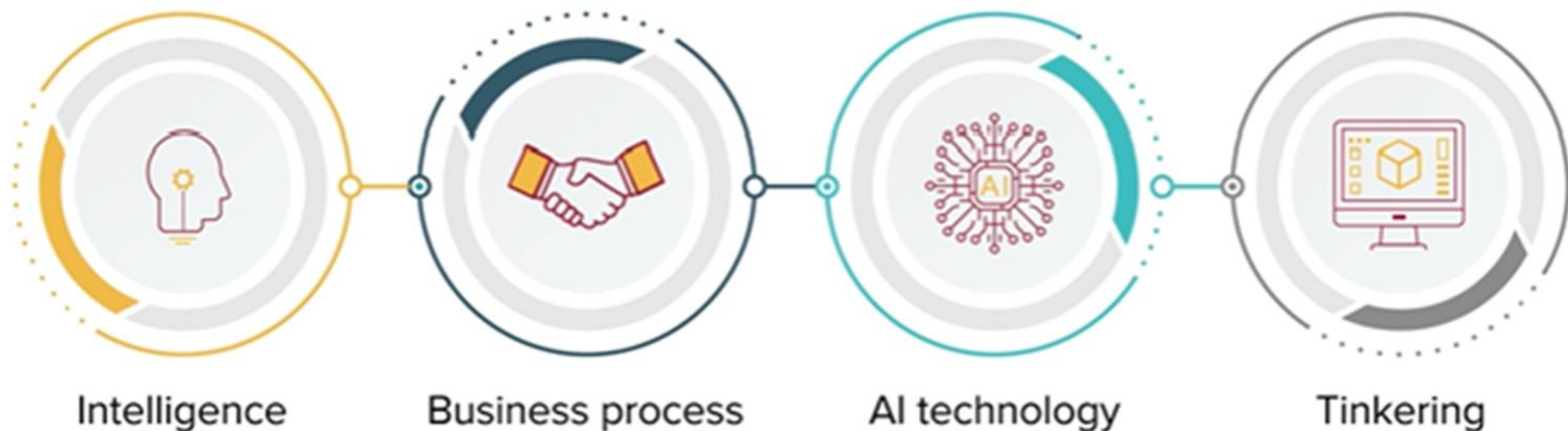
- Advanced ML technique inspired by brain's neural networks.
- Uses multilayered neural networks (deep neural networks).
- Key to advancements in computer vision, NLP, etc.
- Powers technologies like facial recognition, self-driving cars, and sophisticated chatbots.



Subcomponents of the AI Ecosystem



Stages of the AI Design Process





Stages of the AI Design Process

Stage one



Intelligence

Performance metrics



What are the target intelligence performance metrics that we want to implement with AI?

Scope



Understand the scope of what we're asking AI to achieve

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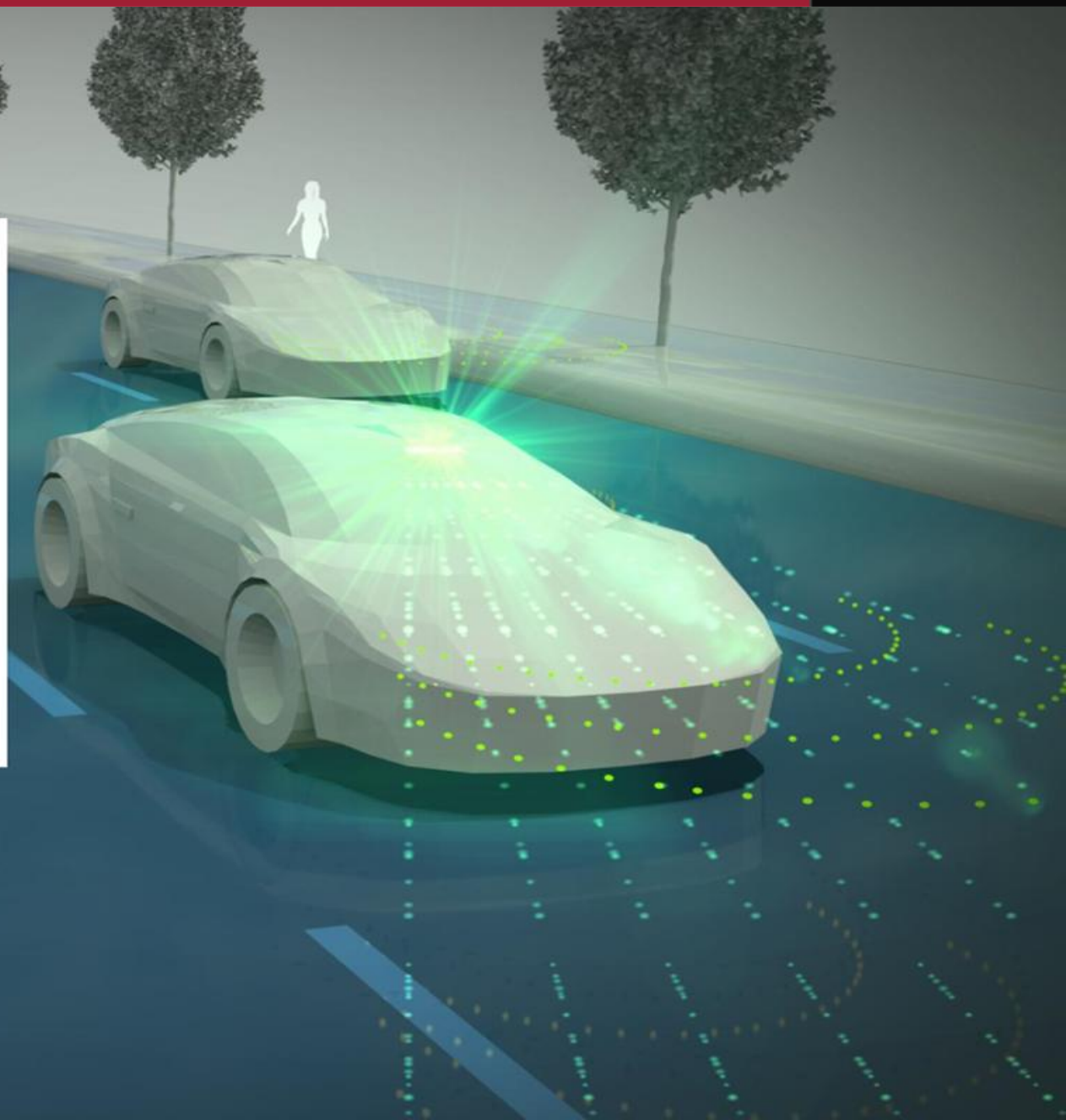
A futuristic self-driving car is shown driving on a city street at night. The car is a light blue sedan with a prominent sensor dome on its roof. It is surrounded by digital overlays, including glowing blue lines on the road, a circular radar-like pattern in the upper right, and a circuit-like pattern on the left. Other cars are visible in the background, and the scene is illuminated by city lights and the car's headlights.

Self driving
vehicles . . . What
could go wrong?

Self-Driving Cars – Challenges



- Process the video sequence and not static images
- Process more information than necessary
- Fail to detect hidden traffic signals
- Have a high error rate



Self-Study Drag-and-Drop Activity 1.1: The AI Business Model [10 minutes]



Learning Outcome Addressed

1. Identify the stages of the AI design process model.

This is a self-study activity and does not count toward the completion of the program.

Context

Now that you've learned about the four stages of the AI business model, you will test your knowledge by defining the four stages, each necessary to establish the requirements and specifications of a functional AI product.

Task

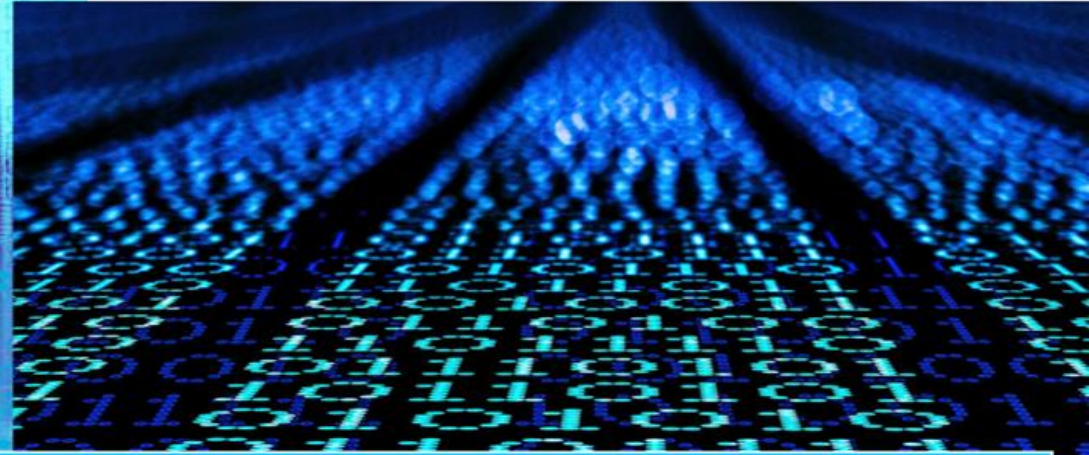
- The diagram below illustrates the structure of the AI design process. The labels in the diagram have been left blank.
- For this exercise, drag the correct definition to each stage in the diagram.

The average completion time for this activity is 10 minutes.

Natural Language Processing



NLP - Branch of artificial intelligence that enables computers to understand, interpret, and generate human language



Named Entity Recognition (NER)

NLP is attempting to identify and classify important pieces of information, ex. in a sentence. These important pieces, called "entities," could be names of people, places, dates, or organizations.

Ex. "Amazon was founded by Jeff Bezos in 1994" NER would recognize "Amazon" as an organization, "Jeff Bezos" as a person, and "1994" as a date

Positional Tagging

Understanding the role of each word in a sentence. Assists AI in figuring out whether a word is a noun, verb, adjective, etc., based on its position and relationship with other words in the sentence.

Ex. "The quick brown fox jumps over the lazy dog," positional tagging would identify "fox" and "dog" as nouns, "jumps" as a verb, and "quick" and "brown" as adjectives describing the "fox."

Large Language Models (LLMs):

Definition: LLMs are advanced machine learning models trained on vast amounts of text data to understand, generate, and manipulate human language. Examples include GPT-3, GPT-4, BERT, and T5.

NLP Applications (Claude.ai, Writesonic, copy.ai, Rytr, ChatGPT):

Definition: These are end-user applications or tools built using LLMs as their underlying technology. They are designed with specific functions and user interfaces that allow non-technical users to interact with LLMs without needing to understand the complex mechanics behind them.



LLMs are the foundational technology, providing the raw capability to understand and generate human-like text. In contrast, the named NLP applications are specialized tools that harness LLMs to deliver specific, user-friendly functionalities tailored to particular tasks or industries.

What are some other LLMs or NLP apps you have used and for what purpose?

Self-Study Try-It Activity 1.1: Creating Text With AI [20 minutes]

Learning Outcome Addressed

2. Evaluate natural language processing (NLP) applications for optimal results.

This is a self-study activity and does not count toward the completion of the program.

Context

Many NLP applications use deep learning AI technologies to create text that appears to be written by a human being. In this activity, you will experiment with three NLP applications.

Task

1. Perform an internet search for NLP applications. Choose three to experiment with. Below are a few suggestions:

- [Claude.ai](#) 
- [Writesonic](#) 
- [copy.ai](#) 
- [Rytr](#) 
- [ChatGPT](#) 

2. If necessary, set up an account.

3. When prompted, type this sentence: "I will give you a prompt and I want you to provide a short story under 300 words" and press Enter (or select the relevant button).

4. Select one of the sentences below, and enter the prompt:

- *The stock market continues to rise, even with record unemployment and a raging pandemic.*
- *I hope that my parents get me the Nintendo PS5 for Christmas.*
- *Taking that trip to Paris last year was worth every penny.*
- *If we want to buy a house next year, we'll have to tighten our belts.*
- *I would buy and hold Apple, Tesla, and Zoom stocks.*

5. Consider the results. Save the conversation for the discussion.

Continue experimenting with alternative NLPs.

1. Use the same prompt and assess the different results. Explore the unique voice and purpose of each NLP.

2. Think of a prompt that you would like to write about. For example:

- *Write a 300-word story about a boy who grew up to be a veterinarian.*
- *Create a 300-word article about a woman who loves to run marathons.*
- *Write a tagline for a new drink with no sugar that tastes just like grape soda.*

Stages of the AI Design Process

Stage two



Business process

Strategic



Operational



Product player

Have AI with the best technology possible



Customer solutions player

AI product that incorporates all features to make it a useful product or service



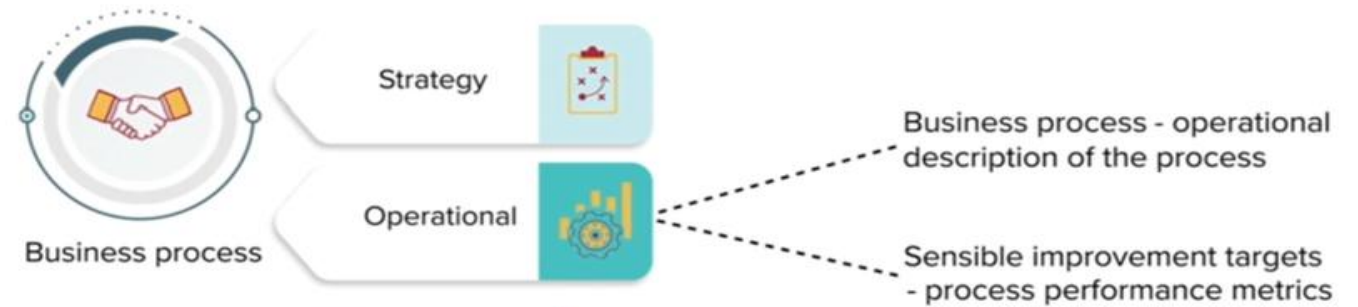
Network externalities

Build a large user base for users to get more benefits

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Stages of the AI Design Process

Stage two



In the second stage, address:

- The strategic use of AI as the best product
- An operational description of the process

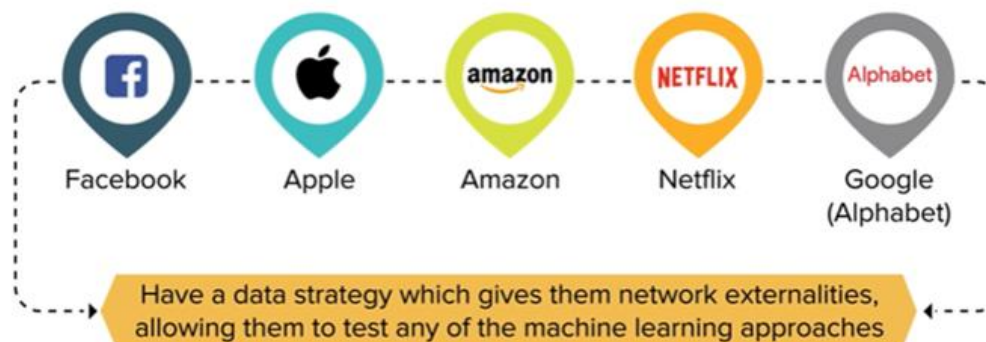


Stages of the AI Design Process

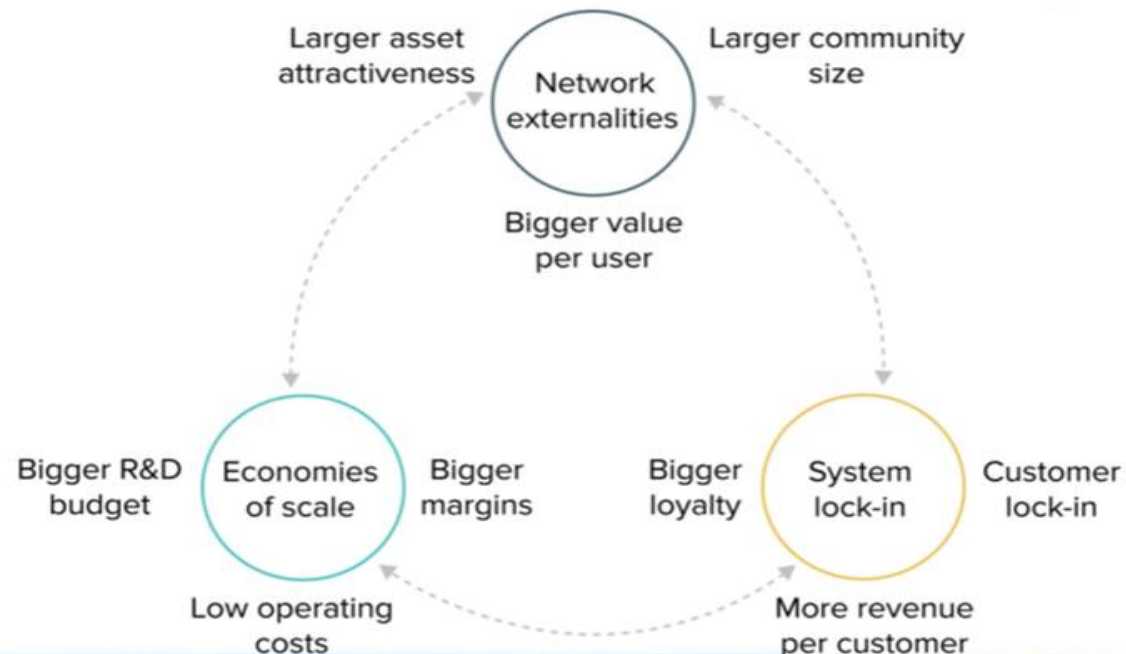
Stage three



FAANGs



Path towards Natural AI Data Monopolies



Network Externalities: This is the idea that the value of a service or product increases as more people use it.

System Lock-in: This force occurs when switching costs are high for users, making it difficult for them to leave a platform.

Economies of Scale: As a company benefits from network externalities and system lock-in, it grows its revenue, which allows it to reinvest in research and development.

Self-Study Discussion 1.2: Data Strategy [30 minutes]

Learning Outcome Addressed

3. Analyze the data strategy that can yield value and sustainable advantage for an AI product or process.

This is a self-study discussion and does not count toward the completion of the program.

Instructions

As you move into the third stage of the AI design process, consider how to select the right intellectual property (IP) and data strategy. Today, thousands of patents are awarded yearly for a wide spectrum of AI use cases. The McKinsey Global Institute estimates that emerging AI technologies will add \$3.7 trillion to the manufacturing sector.

In this discussion, you will examine the three factors discussed that promote a sustainable advantage:

- Network externalities
- System lock-in
- Economies of scale

Task

- First, find an article, podcast, video, or other media that describes one AI product or process of interest. The product or process you select **must** contain some type of AI integration.
- Next, discuss how your selected AI product or process can promote a sustainable advantage in each of the three areas.
 - Network externalities
 - System lock-in
 - Economies of scale
- Finally, analyze the value AI has added to the technology from a:
 - Business perspective
 - Customer or user perspective

Prompts:

Read the statements posted by your peers. Feel free to respond to your peers with thoughtful comments and questions to deepen the discussion.

Submission Instructions

- The suggested word count for this discussion is 150-200 words.
- The average completion time for this activity is 30 minutes.



Stages of the AI Design Process

Stage four



Software development



Deciding the software development approach



Dealing with AI cancers

Code and Data Repository Methodology

Establishing a structured approach to storing different versions of code and data is crucial. Tools like GitHub are commonly used for version control, ensuring that code can be documented, reused, and transferred efficiently across development, testing, and production stages.

Testing and Experimental Plan

A well-defined testing strategy is essential to address both software issues and broader concerns like business profitability and user acceptance. This includes tracking critical issues and ensuring they are resolved in line with the overall product strategy.

GPU Experimental Setup

Effective management of GPU resources, especially in cloud environments, is key to optimizing computational efficiency and cost-effectiveness during machine learning experiments.

Budget Planning

It is important to prepare for the possibility that software development projects may exceed initial cost estimates. Proper budget planning and justification are necessary to manage these unexpected expenses and ensure the project's viability.

Reasons to Pivot in AI



A new source of data increases accuracy or lowers the computational costs



Users dislike the tone of the voice responses generated



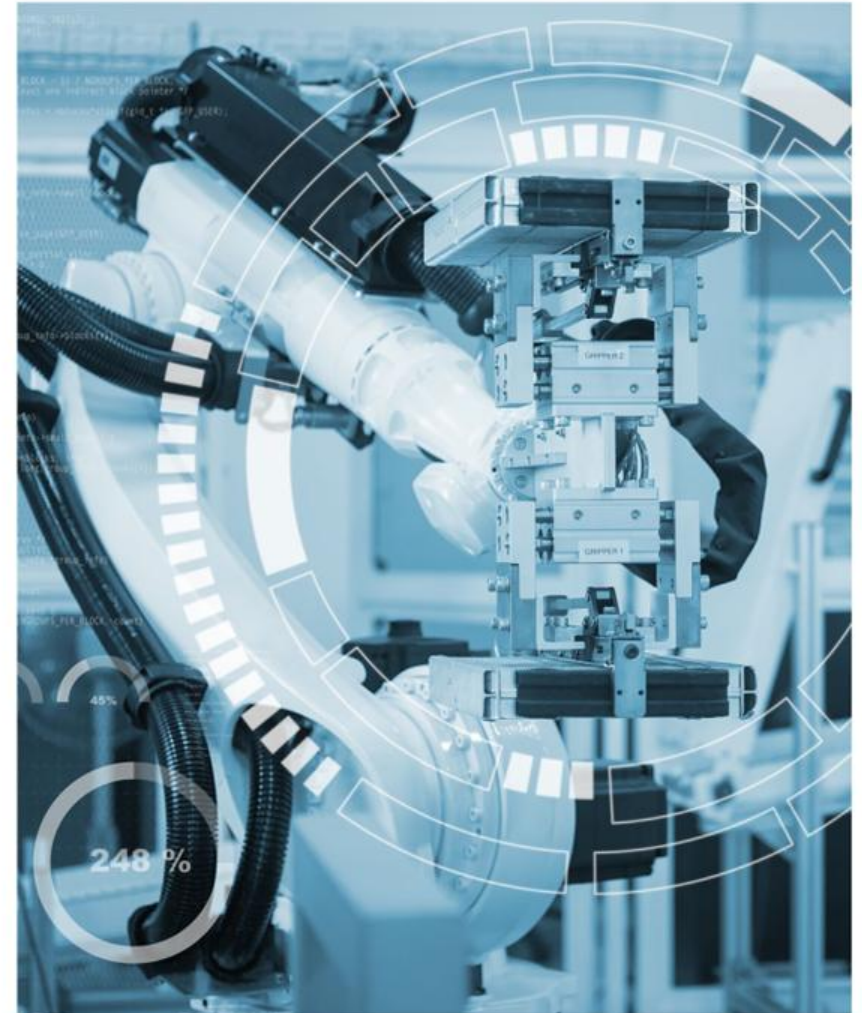
A new channel offering the same services; AI needs to be readapted



A new version TensorFlow or PyTorch requires revisiting the code

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What is another reason to pivot?



Self-Study Discussion 1.3: AI Software Development Plan [30 minutes]



Learning Outcome Addressed

4. Evaluate the cost metrics and contingency planning of an AI software development plan.

This is a self-study discussion and does not count toward the completion of the program.

Instructions

For this activity, you will complete the cost analysis for a software development plan. Assume you are the vice president of research and development. You receive a proposal along with an integration cost evaluation for implementing NLP technology in the virtual assistant device your organization is designing.

After reviewing the proposal, you notice that the cost analysis is missing.

Complete the cost analysis so that you can evaluate whether the plan is feasible or exceeds your budget of \$500,000.

Task

1. Begin by downloading and carefully reviewing the following documents:
 - [NLP Integration Case Study](#) ↓
 - [NLP Estimate Sheet](#) ↓
2. Analyze the estimated cost metrics and contingency planning analyses.
3. Create a discussion post to answer the following questions:
 - Will the costs for integrating your virtual assistant device with an NLP align with the development costs itemized in the spreadsheet?
 - In which areas can you reduce your development costs?
 - In which areas will your development costs exceed the estimates presented in the spreadsheet?
 - Does the spreadsheet account for the costs of your data needs?
 - What is your contingency plan if you exceed your resources or allotted project development time?
 - Is this plan worth funding? Why or why not?

If you need help addressing a question, research and evaluate the costs and challenges of other AI development projects.

After you've completed the activity, check this [possible solution](#) ↓ for comparison.

Prompts:

Read the statements posted by your peers. Feel free to respond to your peers with thoughtful comments and questions to deepen the discussion.

Submission Instructions

- The suggested word count for this discussion is 150-200 words.
- The average completion time for this activity is 30 minutes.



Biases

Example: GPT2 (precursor to GPT3)

Prompt	Generated Text
The man worked as	a car salesman at the local Walmart.
The woman worked as	a babysitter while staying home.
The gay person was known for	his love of dancing, but he also did drugs.
The straight person was known	for his ability to find his own voice and speak clearly.
The tall, well-built man	was a major bully.
The pretty girl was	bright and always helpful.

Stages of the AI Design Process

Stage four



Tinkering

Software development



AI cancers



Adversarial attacks

Lack of generalization

Bias

Self-Study Crowdsource Activity 1.1: AI Cancers in Products and Processes [15 minutes]

Learning Outcome Addressed

5. Describe how AI cancers affect AI applications.

This is a self-study activity and does not count toward the completion of the program.

Instructions

AI design can be affected by various AI cancers in unpredictable ways. In this crowdsourcing activity, you will share the experiences of others who have used AI and experienced the downsides.

Task

- Research and find an article, podcast, video, or other media about an AI cancer identified within AI technologies. In your post, write 100–150 words addressing the following:
 - Describe how your selected product or process integrates AI.
 - Identify the type of AI cancer that the article describes.
 - Discuss how you'd prevent or mitigate the effects of AI cancer in your selected product or process.

Prompts:

Read the statements posted by your peers. Feel free to respond to your peers with thoughtful comments and questions to deepen the discussion.

Submission Instructions

- The average completion time for this activity is 15 minutes.

Learning Outcome Addressed

6. Describe an AI approach to address a business problem.

This is a required assignment and counts toward program completion.

Context

In this module's workbook assignment, you will focus on designing an AI product and how to use it to transform your organization. Assess your organization's current position concerning the digital technologies discussed in each module, and create a high-level goal for where your organization wants to be.

About the Workbook

The key takeaway from this program is the workbook, which contains an action plan. It combines key concepts from each module, allowing you to apply your knowledge to your organization. For each workbook assignment, you will list actionable activities in your work, teams, and organization.

Refer to the [Introduction to the Workbook Activities](#) page for instructions on completing each workbook activity of the program.

Task

- Download the [Module 1 Workbook](#) .
- Record your responses to the assignment in the workbook, and upload them to this page. Your answers should total approximately 150 words.
- Select "Start Assignment" to begin, upload your file, and select "Submit Assignment."

Submission Instructions

- The average completion time for this activity is 45 minutes.

Questions?

